

Energy Efficiency and Solar Electricity Go Hand in Hand

by Steve Vang and Rob Hammon

While zero-energy homes (ZEHs) are still more of a goal than a reality, near-zero-energy homes—those that produce as much electricity, but not quite as much total energy, as the residents consume—are up for sale. Indeed, about 12 cutting-edge builders have constructed whole communities of these near-zero-energy homes (NZEHs). These communities of NZEHs have attracted widespread media attention and, more importantly to builders, have sold either better than or as well as their more conventionally built counterparts. These visionary builders have learned some important lessons about how to build and sell homes that are practically energy independent.

DOE, through its Building America (BA) program, has been collaborating with these forward-thinking builders, as together they work toward DOE's 2020 goal of creating marketable homes that are highly energy efficient and that produce as much energy as they use. BA research focuses primarily on developing and integrating leading-edge technologies into ZEHs, on developing rate structures to encourage reduction of electricity demand during the summer peak periods, and on developing controls to assist or automate reduction of peak demand.

There are currently six BA teams in the United States. Each team is responsible for developing practical and affordable approaches to increasing the energy efficiency of new homes in one or more of the six U.S. climate zones. The Building Industry



DAVID MYBERG

Benefits to the builders of near-zero-energy homes range from cost savings from rebates to greater experience with innovative building technologies.

Research Alliance (BIRA) team, led by ConSol—the company we work for—is responsible for the marine and hot-dry climate zones, which include parts of California. BIRA's NZEH projects involve multiple builders, locations, and utilities in California. These projects provide a rich array of input on the advantages and disadvantages of building NZEHs, and on the lessons to be learned from doing so.

Currently, the closest thing to zero-energy homes are the homes in these NZEH communities. While a home can achieve a zero-energy bill by building with today's technology, it is too expensive and builders are

understandably reluctant to take this approach. BIRA's current goal for NZEH communities is to reduce each home's total energy bill by at least 60%, and for each home to produce as much electricity as it consumes annually.

All BA program teams use a systems engineering approach to produce homes on a community scale that

- use 40%–70% less energy than homes built to the BA benchmark (nationally);
- are constructed in less time than standard-built homes, and with less waste;

- are more comfortable than standard-built homes;
- are more durable than standard-built homes; and
- are more marketable than standard-built homes.

In order to reach the goal of near-zero energy, most builders think that PV panels must be used to offset at least some of the home's electricity consumption. But because PV systems are expensive, it is imperative to first build a super-energy-efficient home. This reduces the amount of electricity needed, which in turn makes it possible to install a smaller PV system. NZEHs incorporate super-energy-efficient building features and a properly sized PV system. Some of those superefficient features are described below.

High ceiling and wall insulation. Higher insulation levels—for example, R-38 or even R-49—help keep heat out of the house during the summer and maintain heat inside the house during the winter.

Radiant barrier. Installed on the inside of the roof to reduce solar heat gain and keep the attic cool, radiant barriers are important in California, where heating and cooling ducts typically run through the attic.

Low air infiltration rate. Tight building envelopes help to reduce air flow into and out of the home. With lower infiltration of unconditioned air and exfiltration of conditioned air, less conditioning is required, making the home more efficient. Homes with low air infiltration and exfiltration are often quieter and cleaner.

High-performance windows with spectrally selective glass. High-performance glazing increases the comfort of the home and reduces summer cooling requirements by reducing solar heat gain into the house. Spectrally selective glass lets in most of the visible sunlight while blocking approximately 80% of the infrared and ultraviolet solar energy that drives up cooling costs and degrades curtains, carpets, and furniture. In the winter, this glass helps to reduce

heating costs by reflecting room-side radiant heat back into the room.

Engineered HVAC. Licensed mechanical engineers size and select HVAC systems, based on ACCA *Manual J, S, and D*. The final product is a properly designed HVAC system with correct duct sizes and registers placed to ensure that conditioned air will be evenly distributed throughout the whole house. These systems improve both energy efficiency and comfort.

Tight duct installation. Tight installation eliminates excessive air leakage in the duct system. Excessive air leakage makes the HVAC system work harder, using more time and energy to cool and heat the home. Duct leakage can also increase indoor moisture levels and air flow in and out of the house. It can also make the house drafty and uncomfortable to live in.

Tankless water heater. Tankless water heaters typically have high efficiencies, with energy factors, or EFs, of 0.80 or higher. These heaters have no storage tank; the water is heated on demand, which avoids the energy losses associated with reheating the water that is sitting in storage.

All-fluorescent lighting. A fluorescent lightbulb uses about 75% less energy and lasts up to 10 times longer than an incandescent bulb. Properly specified fluorescents produce high-quality, efficient lighting.

PV (solar-electric) system. A PV system generates electricity when the sun shines on the PV panels. The size of the PV system depends on the location, efficiency, and size of the home.

The features listed above are typical of NZEH projects in California and Nevada. However, to maximize energy savings, some of these projects are incorporating more innovative technologies (see “Innovations at Hand”).

BENEFITS TO THE BUILDER

We found that building NZEHs or NZEH communities provided five benefits to the builders that ranged from cost savings through rebates and incentives to greater experience with new building technologies, including PV. These benefits are discussed below.

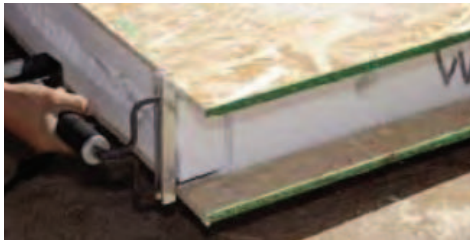
Incentives and Rebates

Builders of NZEHs benefited from federal, state, and utility incentives. These incentives helped to offset the substantial incremental cost of construction. This cost ranged from \$18,000 to \$25,000, or roughly 3%–7% of the cost of the home. After the incentives and rebates, the net incremental cost to the builder ranged from \$8,000 to \$15,000, depending on the size of the house and the climate zone. Obviously, the increased cost of building a NZEH weighs more heavily on the builders of less expensive homes than on the builders of high-end homes. Nonetheless, some of the communities include entry level or below-market, affordable homes. Although incentives and rebates reduce the builder's incremental cost by roughly 50%, that incremental cost is still a significant factor to consider when builders are building NZEHs on a community level. However, in all these NZEH projects, the builders successfully recovered their costs and enhanced their reputations as visionaries and leaders. Clarum Homes has stated that from now on it will build to nothing less than NZEH standards, and John Ralston, vice president of sales and marketing for Premier Homes, has said that “it was worth it” to build to those standards. A NZEH is also “worth it” for the buyer. Well-designed NZEHs produce a net positive cash flow, based on financial benefit analyses, because the savings on the energy bills are greater than the increase in the mortgage payment.

Unfortunately, these rebates and incentives came with a downside: Substantial new paperwork had to

INNOVATIONS AT HAND

Innovations that some near-zero-energy homes feature include the following:



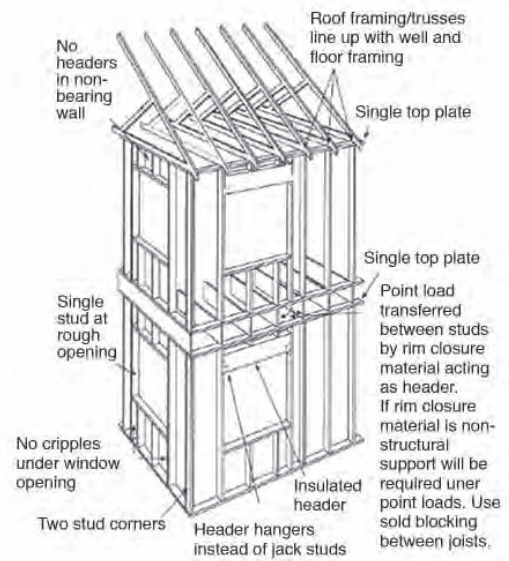
SIP Wall System. The structural insulated panel (SIP) wall system uses a high mass wall (up to R-40 equivalent) to reduce heat gain and loss. Here, a SIP installer trims the bottom of a wall panel. The panel will be joined to other panels side by side with splines to create a continuous thermal barrier.



OASYS Two Stage Evaporative Cooler. This advanced cooling system delivers more capacity to cool the house and has an energy efficiency ratio (EER)-equivalent ranging from 40 to 135.



DOW's T-Mass Wall System. This system has high mass walls (up to R-40 equivalent) that are made from concrete and Styro-foam layers to reduce heat gain into the house during summer months.



OVE Wall System. An optimum value engineering (OVE) wall system has higher wall insulation R-value (up to R-21 equivalent) compared to a traditional framed wall.



NightBreeze System. The NightBreeze system cools the house when the outdoor air temperature is cooler than the indoor air temperature. This helps to reduce the need for the air conditioning system (see Figure D).

Freus A/C. Unlike conventional air-cooled systems, the Freus A/C system uses water to cool the coil of the condensing unit. Freus delivers more capacity and is more efficient in hot weather (up to 18 seasonal energy efficiency ratio (SEER) and 18 energy efficiency ratio (EER)).

Solar Water Heating. Solar water heating helps save gas or electricity consumption by preheating the water in the water heater. During the summer months in Northern California, a solar water heater usually can supply all of a household's needs for domestic hot water.

be generated to acquire them. The paperwork burden was exacerbated by the fact that the rebates and incentives sometimes came from different sources, each with its own documentation requirements. Since builders were already overwhelmed with the new technical details of NZEHs, they did not want to spend the time and effort to collect all of the different rebates, especially with more federal rebates now available. The energy consultants, HVAC contractors, and utilities sometimes helped out, making sure that the paperwork for the rebates was correctly filed, and that the rebates were delivered directly to builders. In the nearterm, the team members mentioned above will need to continue this role and file all rebate paperwork for the builder.

Marketing and Media Exposure

The NZEH projects that we collaborated on are all successful. Thanks to marketing support from the NZEH team, the NZEH projects generally have drawn substantial media and newspaper coverage and have also appeared in numerous trade journals. This media coverage provided tremendous exposure for the projects at little or no cost to the builder, translating into better-than-free advertising. It is difficult to assign a dollar value to the amount of press received by each NZEH project, but John Ralston says that for the Premier Gardens project in Sacramento, California, it was “certainly in the hundreds of thousands, maybe more”—adding that he doesn’t “know much about the cost of advertising on some of these national media, like CBS and Good Morning America.”

Competitive Sales and Product Differentiation

NZEH homes sold as well as or better than competing non-NZEH homes in all of the NZEH projects. For instance, both Premier’s and Clarum’s NZEH communities sold faster than neighboring communities. In the Centex Homes’ Avignon

project in Pleasanton, California, these NZEHs are outselling their competitors three to one, according to Centex Homes. Although the housing market is down this year, Clarum, Grupe Company, and Premier Homes are continuing to build NZEH homes.

Collaborative Partnerships

The NZEH builders increased their involvement with nonprofit, business, and governmental organizations by building near-zero-energy homes. Collaborating organizations included ConSol, utilities, building departments, DOE’s Building America and Zero-Energy Home programs, the National Renewable Energy Lab (NREL), the California Energy Commission (CEC), PV manufacturers, and other product representatives. These partners assisted with the incorporation of energy efficiency features and practices into the homes. ConSol also provided best-practices construction protocols and installer training—especially for insulation and HVAC subcontractors—and, by conducting inspections and tests during construction, ensured that the homes were built to high specifications.

Experience with PV

One of the lessons we all learned from these NZEH developments is that builders can optimally site PV, even when this means that the PV is fully visible from the street. In the past, builders have avoided installing PVs in front of homes for fear that buyers will find them obtrusive and unattractive. Installing a building-integrated PV (BIPV) system at the front of the house to maximize electricity generation was a new tactic. This tactic was first tried at Premier Gardens (see “Sacramento’s Zero-Energy Community,” *HE* March/April ’06, p. 24). John Ralston says that Premier Homes “did not know if this would have a negative effect on the homeowners.” Only after the homes were sold and occupied did Premier find that “the homeowners did not care much, because the building-integrated

PV system blended in aesthetically with the roof tile.” This was great news for Premier, because it can now feel confident installing BIPV on the front of the house to optimize electricity generation. (See “Designing for Solar,” p. 18.)

BUILDER CHALLENGES

Although building NZEHs provided numerous benefits to the builders, it also had some disadvantages, and poor strategies were sometimes employed to sell the homes.

Costs

Even after rebates and incentives, the typical net incremental cost of building a NZEH ranges from \$8,000 to \$15,000—an up-front cost that is economically challenging. The PV system was the most expensive component; it cost approximately \$8,500/kW DC (before incentives). Although the incremental cost of the energy efficiency measures (typically \$2,500–\$4,000) may seem low compared to the cost of the PV, it represents a significant cost to the builder when applied to a community. Builders usually try to cut costs as much as possible and the NZEH package represented a considerable initial risk for builders, since they did not know whether the homes would sell at a price that would fully cover the additional developer costs.

Sales Strategy

Builders build and market NZEHs in three ways: as an option, leaving the sales agent to sell the benefits of NZEH; by building a designated number of preplotted lots to NZEH standards; or by building the entire community to NZEH standards. Our research revealed that it was better to preplot a significant percentage of homes in the community as NZEH or to build a 100% NZEH community than it was to offer a NZEH package as an optional upgrade. When homes were preplotted for NZEH, or when 100% of the homes were



One of the lessons learned from the builders of NZEH developments was that homebuyers were not scared off by building-integrated PV systems, even those installed at the front of their houses.

NZEHs, they sold just as well as non-NZEH homes. This strategy also reduces the cost of each NZEH because the high-efficiency features and PV systems are purchased at volume, and it simplifies field issues because most or all the homes are built in the same manner, with the same features.

However, when NZEH was sold as an option, builders sold one, two, or none of these homes. This was, first, because most home buyers would not spend around \$30,000 (including markups from the builder) to get a NZEH option in a home; and second, because this approach leaves the sale of the upgrade to the NZEH option to a sales agent, most of whom were not trained well enough to explain—or sell—the concept of NZEH. Indeed, we found that the builders' sales staff typically needed several refresher training sessions before they could understand and effectively sell the NZEH concept. Sales agent training is also affected by turnover; trained agents may move on, leaving untrained agents in their place. It is a waste of time, effort and money to the builders, engineers, and PV manufacturers to design homes to NZEH standards when the NZEH option would not sell.

LEARNING CURVES

Sales staff were not the only ones who needed to learn new skills. There were learning curves for both the builders and the subcontractors. The electrician and the roofer had to work together and communicate extensively in order to install the PV systems on schedule. This rarely occurs in typical home construction. The roofer had to lay the PV arrays on the roof and drill a hole through the roof to drop the PV connection wires into the attic. The electrician then had to connect these wires and route them to the inverter and the main electrical panel. This was beyond the normal scope of work for either trade, and it meant that they both had to understand PV installation procedures. Installing the PV systems was also time consuming, since builders had to schedule the arrival and installation of the PV panels.

Getting Approvals

Education was also the key to winning over city plan checkers and inspectors, who sometimes had trouble understanding, and therefore approving, NZEH features. In Premier Gardens, for example, and also in Lakeside, by Morrison Homes, the

duct runs were buried in the attic insulation to improve the efficiency of the HVAC system. In order to do this, it was necessary to lay the ducts on the roof trusses. The city inspectors objected to this procedure. They wanted documentation from the Uniform Mechanical Code (UMC) proving that the procedure was acceptable. City plan checkers and inspectors also questioned PV construction documents as well as PV installation procedures. In the first homes, it sometimes took two to three days to resolve these problems, which delayed the construction schedule. City inspectors also delayed the Clarum Homes Borrego Springs project. This project was on hold for four months because it incorporated high-mass walls and innovative HVAC systems, and the inspectors needed to confirm that these systems met the structural and mechanical codes.

Lack of Turnkey Systems

In a perfect world, builders could implement a turnkey system for constructing NZEHs. Such a system would include a package of energy efficiency measures and a PV system, along with simplified bidding, contracting, permitting, installation, commissioning, maintenance, utility connection, and rebate processing. At present, there are no turnkey systems for NZEHs. However, ConSol is actively working with solar PV providers to provide a turnkey system for the builders that will include energy efficiency measures and a PV system, at a minimum. A full turnkey system would establish a single point of accountability for home builders and their customers. It would minimize the impact of installing new technologies on the builder's construction schedule and reduce a builder's risk in selling the NZEH concept. We see the establishment of turnkey systems as a key element in market transformation.

SAVINGS MEET EXPECTATIONS

We have learned important lessons from each and every NZEH community. Not surprisingly, the electric bills confirm that NZEHs use less energy. Based on the electric bills analyzed by the Sacramento Municipal Utility District (SMUD), the homes at Premier Gardens use less than half the annual electric energy used by comparable homes built to Title 24 standards. More importantly to SMUD, these homes also help reduce districtwide peak demand for summer afternoon loads. The afternoon PV generation overlaps with the utility's peak load period. The Premier Gardens homes averaged a greater than 50% reduction in peak electricity use during the summer of 2005. This directly benefits the utility by reducing peak demand and indirectly benefits the ratepayer, who ultimately pays for new infrastructure needed to meet growing peak demand, were it not offset by the NZEH.

For individual homes, energy savings depended on the residents' lifestyle. Some NZEHs used more energy than typical Title 24 homes do. For example, in one of the NZEH projects, one of the homes had consistently low energy bills for a few months; the energy bills suddenly tripled when the residents installed a spa and a swimming pool. On average, though, NZEHs use substantially less energy than typical Title 24 homes. We found that NZEH homeowners were very pleased that they would be saving money on their energy bills, and that they would be cushioned against energy price volatility in the future.

OVERCOMING MARKET BARRIERS

ConSol and its partners are working on a marketing message for the builder that will help convey the value of NZEHs to the mainstream home buyer. As part of this effort, they are reviewing and critiquing grand openings and press events

at NZEH projects to garner more information. Findings from recent NZEH projects suggest the following:

1. Anecdotal findings to date indicate that NZEH communities sell as fast as, or faster than, competing communities of conventional homes. As more NZEH communities are developed, market absorption data should be collected to confirm or refute these findings. An example is the Grupe Company's Carsten Crossings project, in Roseville, California. Customers are provided with financial benefit analyses comparing a standard Title 24 home, a home with energy efficiency features only, and the as-offered NZEH (with the energy efficiency features and solar). The purpose is to show that the buyer of a NZEH will enjoy a net positive cash flow from the first month of occupancy, because the dollar amount saved on the energy bill is greater than the increase in mortgage cost to finance the NZEH incremental costs. Since the project opened, Grupe has sold twice as many homes as its competitors, probably because it has a knowledgeable sales staff, technical support from the PV manufacturer, and a positive financial benefit analysis.

2. NZEHs are less expensive to build when the features are standard, rather than being offered as an option, due to economies of scale and production scheduling.

3. NZEHs will sell well only when the sales staff are knowledgeable and well trained. Initial in-depth sales training should be augmented with regular—perhaps monthly—mini-trainings, to ensure that the agents are comfortable with, and competent to explain, NZEH information. A sales agent hotline would be beneficial for agents who have buyers with questions that they cannot answer immediately. These questions should be included in future in-depth trainings.

4. The NZEH homeowner often becomes a NZEH cheerleader after receiving a number of very low energy bills. Builders should begin

enlisting current NZEH homeowners to be on hand at grand openings to talk to prospective buyers about their energy bills and about their own comfortable homes. These homeowners can fuel the fever better than any brochure. They could be offered a dinner coupon for their efforts.

Moving the NZEH market from early adopters to mainstream buyers poses many challenges. But the bottom line is that NZEHs are beneficial to the builders, the consumers, the utilities, and the environment. NZEHs sell well and allow builders to differentiate themselves from their competitors. For homeowners, NZEHs produce a positive cash flow from day one, because the energy savings exceed the increased mortgage cost. These homeowners are insulated against higher electricity rates and they are getting a higher-value home. NZEHs significantly reduce summer peak demand—an important utility goal. And they benefit the environment by dramatically reducing CO₂, sulfur dioxide, and nitrogen oxide. As John Ralston so aptly put it, it's worth it to build near-zero-energy homes.



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To learn more about the Building America program, go to www.buildingamerica.gov or www.bira.ws.

For information about the NZEH projects by Clarum Homes and Premier Homes go to: www.clarum.com or www.builtbypremier.com.